ITAI-1371-Intro to Machine Learning-RT-14274

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Lab 06 Reflective Journal

Regression and Classification

This lab helped me understand the difference between regression and classification in a much more practical way. Before working through the exercise, I understood that both are types of supervised learning, but I did not fully see how the same dataset could be used for two completely different machine learning problems. Using the Titanic dataset made this distinction much clearer because we used the same general data source but changed the target variable and the type of model depending on what we wanted to predict.

For the regression portion of the lab, the objective was to predict a passenger's Fare using Age and Pclass as features. Because Fare is a continuous numerical value, Linear Regression was appropriate for this problem. One concept that became clearer to me was that the model is not simply finding an exact answer for every passenger. Instead, it learns relationships between the input features and the target and then uses those relationships to estimate a value for new data.

The regression model produced a Mean Squared Error of approximately 3364.92. Looking only at this number was initially difficult for me to interpret because squared error does not use the same units as the original Fare values. Calculating the square root of the MSE gave an RMSE of approximately 58.01, which made the result easier to understand because it is expressed in the same general units as Fare. This helped me understand why selecting the correct evaluation metric is important. A metric is useful only if we understand what it tells us about the model.

Another concept I learned from Linear Regression was the meaning of the model coefficients. The .coef_ values represent how each feature influences the model's prediction while the other features are held constant. This helped me see that training a model means the algorithm is actually learning numerical relationships from the training data rather than following manually programmed rules.

The classification portion made the difference between the two types of supervised learning even clearer. In this part, the objective was to predict whether a passenger Survived using Age, Pclass, and Sex. Since the possible result is a category, either survived or did not survive, this became a classification problem. Logistic Regression was used even though its name contains the word “regression.” I found this important because the name can initially be confusing. Logistic Regression is used here to estimate the probability that an observation belongs to a particular class and then make a classification.

The Logistic Regression model achieved an accuracy of approximately 74.83%. Accuracy makes sense for this problem because the predictions can be compared directly with the correct class labels. This also helped me understand why accuracy would not work for predicting Fare. A predicted Fare of $50 compared with an actual Fare of $52 is not simply “correct” or “incorrect”; we need to measure how far the prediction is from the actual value. In classification, however, we can determine whether the predicted category matches the actual category.

One of the strongest connections I made in this lab was with the previous work we have done with training and testing data. In both problems, the dataset was divided into training and testing sets before the model was trained. The training data is used by the model to learn patterns, while the testing data provides information that the model has not used during training. This separation is important because evaluating a model on the same data that it learned from would not give us a realistic idea of how it performs on new information.

I also noticed how preprocessing connects directly with model building. For example, rows with missing Age values were removed, and the Sex feature was converted from text categories into numerical values before it could be used in the classification model. This reinforced what we learned in the previous labs about data preparation. Cleaning and transforming data are not separate activities from machine learning; they are part of the complete workflow needed before a model can make useful predictions.

Overall, this lab helped me move from simply understanding definitions to seeing the complete supervised learning workflow: select the features and target, split the data, create the appropriate model, train it, make predictions, and evaluate the results using a metric that matches the type of problem. I now have a clearer understanding of how the target variable determines whether a problem should be treated as regression or classification. This will also be useful for our Real Estate project because predicting a continuous value such as Sale Amount would require a regression approach rather than classification.